

Proposal Review Report

February 12, 2026

Proposal Summary

The proposed research addresses a persistent concern in psychological science: whether preregistration and Registered Reports restrict exploratory research, thereby constraining scientific creativity and discovery. Since psychology began adopting these practices approximately 15 years ago to address replication failures and questionable research practices, critics have argued that requiring advance specification of hypotheses and analyses would reduce valuable exploratory work. Proponents counter that registration merely clarifies the distinction between confirmatory and exploratory analyses without restricting either. This study will examine 300 articles across four categories: Registered Reports ($n=75$), preregistered articles ($n=75$), matched traditional articles from the same journals and timeframe ($n=75$), and traditional articles from 2010 pre-dating registration adoption ($n=75$). Trained raters will code each article for the presence of reported exploratory research, defined as any analysis not indicated as a planned test of a specific hypothesis, across three manuscript sections (Introduction, Results, Discussion). The authors hypothesize that registered articles will report more exploratory research than traditional articles, reflecting improved transparency rather than restriction. They further predict Registered Reports will show greater reporting than preregistered articles due to more stringent review processes. The study aims to provide empirical evidence addressing claims about registration's impact, potentially informing ongoing debates about implementing these practices across psychological science and other disciplines adopting similar transparency initiatives.

As part of our partnership with Lifecycle Journal, pre-registration submissions undergo an independent evaluation through Paper-Wizard's AI-powered review system. This report provides structured feedback to support research planning and refinement. This review was generated by Paper-Wizard's automated analysis engine and is provided directly to authors as part of the evaluation process. Lifecycle Journal does not modify or influence the content of this review.

General Review Synopsis

This proposal addresses a timely metascience question but requires fundamental methodological refinement to yield valid inferences. Three critical issues warrant attention. First, the operationalization measures reported exploratory research rather than its actual conduct, creating a construct validity problem that prevents adjudicating whether registration restricts exploratory work or merely clarifies its reporting. This distinction is central to addressing critics' concerns yet remains unresolved. Second, systematic measurement bias arises from detecting only explicitly labeled exploratory analyses—traditional articles likely contain unlabeled exploratory work, while registered articles face explicit labeling requirements, artificially inflating differences independent of actual practice changes. Third, the statistical approach requires revision: chi-square tests violate independence assumptions given journal-level clustering from the matching procedure, no corrections address multiple comparisons across 4-6 hypothesis tests, and the 2010 comparison group introduces temporal confounding that cannot be separated from registration effects. Addressing these through construct refinement, two-stage coding distinguishing labeled versus unlabeled exploration, mixed-effects models accounting for clustering, and temporal controls would substantially strengthen the study's capacity to provide definitive evidence for this important debate.

Novelty Assessment

Novel - The proposal presents empirical novelty by systematically testing a long-debated claim that preregistration/registered reports restrict exploratory research. Despite extensive searches, no published papers were found that directly compare the frequency of reported exploratory analyses across registered vs. traditional articles in psychology. The authors cite O'Mahony (2023) as the only relevant data, but this is an unpublished PhD thesis (<https://orca.cardiff.ac.uk/id/eprint/167686>) that examined this question within a larger project. The current proposal substantially extends this by: 1) including preregistered articles (not just registered reports), 2) adding historical comparison (2010 articles), 3) using systematic matching criteria, 4) employing transparent coding procedures with reliability assessment, and 5) explicitly testing temporal trends. The proposal provides the first peer-reviewed, systematic empirical test of whether registration practices actually suppress exploratory research reporting as critics claim, moving the debate from opinion to data. The research addresses a scientifically and practically important question about open science reform impacts with a rigorous design that will provide definitive empirical evidence currently absent from the published literature. For details on the most similar publications, see Appendix 1.

Issues

1.1 Construct Validity and Operationalization

Alignment Between Measurement and Research Question -

Critique: The study's core limitation—measuring *reported* exploratory research rather than its actual conduct—requires careful consideration regarding whether the design can meaningfully address critics' concerns about registration *restricting* exploratory work (Nosek, 2018)

(Goldin-Meadow, 2016). Critics argue registration constrains research practices ("putting researchers in chains," "stifling discovery"), yet the operationalization conflates reporting transparency with behavioral impact. Observing Pattern 2 (more reported exploration in registered articles) could reflect improved labeling rather than unrestricted conduct, while Pattern 1 might indicate restriction or merely poorer reporting. This ambiguity prevents adjudicating between registration enabling exploration, clarifying it, or restricting it while enhancing visibility of remaining work.

Relevant section: Abstract; "The Present Study" section, particularly the acknowledgment that "the current study will not be able to actually assess the frequency of exploratory research, per se, but rather the frequency with which exploratory research is reported and transparently described" (p.5).

Framework for Addressing: - Reframe research questions explicitly in terms of reported exploratory research transparency rather than actual conduct restriction (e.g., "Is exploratory research more transparently reported in registered versus traditional articles?")

- Add a dedicated limitations subsection explicitly discussing how findings about reporting practices do/do not address critics' concerns about practice restriction

- Develop supplementary analyses that could more directly assess conduct (e.g., examining preregistration-manuscript discrepancies to identify unreported exploratory work in registered articles)

- Specify in the Discussion how different patterns would be interpreted: finding more reported exploration in registered articles supports transparency claims but does not refute restriction of actual conduct

- Consider adding author surveys or interviews with a subsample to directly assess whether registration affected their exploratory research behavior versus only their reporting

Systematic Detection Bias in Traditional Articles -

Critique: The operational definition introduces measurement error systematically favoring the hypothesis by detecting only explicitly labeled exploratory research. Traditional articles written before widespread preregistration norms (particularly 2010 articles) likely contain unlabeled exploratory analyses because authors had no reason to distinguish confirmatory from exploratory work (Wagenmakers et al., 2012). Conversely, registered articles face explicit requirements (Registered Report formats mandate separate labeling; preregistration badges require this distinction) to flag exploratory work. This creates artifactual differences: registered articles will have exploratory work explicitly labeled per journal requirements, while traditional articles may contain equivalent exploration remaining undetected. The study cannot distinguish whether registered articles contain MORE exploratory work or simply LABEL it more, directly affecting the ability to test competing explanations.

Relevant section: Method section, "Statements of Exploratory Research" coding criteria (pp.10-11).

Framework for Addressing: - Implement a two-stage coding process: first, identify explicitly labeled exploratory research; second, conduct detailed analysis plan comparisons (for registered articles with accessible registrations) to identify conducted-but-unlabeled exploratory analyses

- Add systematic searches for common exploratory analyses that may be unlabeled in traditional articles (e.g., post-hoc subgroup analyses, alternative model specifications, additional dependent

variables not mentioned in introduction)

- Code separately: (a) explicitly labeled exploratory research and (b) analyses meeting the definitional criteria but unlabeled, then analyze these as distinct outcomes
- Include sensitivity analyses comparing only registered articles with accessible, detailed preregistrations to traditional articles, where unlabeled exploration can be more reliably detected through plan-manuscript comparison
- Report both "any exploratory research present" and "exploratory research explicitly labeled as such" as separate outcomes to distinguish transparency from conduct

Operationalization of Context-Dependent Exclusions -

Critique: The protocol's reliance on subjective "context" for excluding certain analyses (interaction probing, exploratory factor analysis) introduces inconsistency without objective specification. The authors cite (Baranger et al., 2023) acknowledging interaction hypotheses typically lack specificity, yet exclude interaction probing when it "follows a hypothesized interaction"—contradicting recognition that non-specific interaction predictions render follow-up probing exploratory by the study's definition ("any analysis not a planned test of a specific hypothesis"). Without objective criteria, raters may systematically judge interaction probing as confirmatory in registered articles (where interactions were preregistered) but exploratory in traditional articles, even when hypothesis specificity is equivalent.

Relevant section: "Statements of Exploratory Research" (pp.10-11), particularly the paragraph discussing "not all uses of the word 'explore' or 'exploratory' necessarily indicate exploratory research."

Framework for Addressing: - Provide explicit, objective criteria for context-dependent decisions: "Interaction probing is exploratory unless (a) the preregistration/introduction specifies the directional pattern of simple effects, AND (b) the analysis matches this specification exactly"

- Develop detailed decision trees with specific examples for each ambiguous category (interaction probing, EFA, robustness tests) showing when each would/would not be coded as exploratory
- Include these decision rules in the codebook provided during training with 5-10 worked examples per category
- Track and report separately the frequency of context-dependent decisions by article type to assess whether systematic biases emerge
- Consider including all interaction probing consistently as exploratory unless extraordinarily specific a priori predictions are documented, given Baranger et al.'s demonstration of typical non-specificity

Construct Representation Through Binary Coding -

Critique: Binary presence/absence coding across three sections, averaged to create 0-1 scores, severely underrepresents the construct and cannot detect meaningful restriction. Two articles receiving identical scores could differ dramatically: one reporting a single footnote robustness check versus another with extensive exploratory analyses comprising 50% of Results driving primary conclusions. If critics' concerns about "putting researchers in chains" have merit, effects would likely manifest in reduced proportion of results devoted to exploration, less prominent placement, fewer exploratory findings driving conclusions, or reduced sophistication—dimensions this coding cannot capture. The sensitivity analysis suggesting detection of $d=0.35$ misleadingly assumes binary coding adequately represents the construct,

when meaningful restriction could exist with "present" exploration that is perfunctory or minimized.

Relevant section: "Article Coding" section (pp.9-11), particularly "Statements of Exploratory Research."

Framework for Addressing: - Expand coding to capture: (1) count of distinct exploratory analyses reported, (2) estimated proportion of Results section text devoted to exploratory findings (categorical: 0%, 1-10%, 11-25%, 26-50%, >50%), (3) whether exploratory findings contribute to primary conclusions (yes/no), (4) prominence of exploratory work (main text vs. supplement/footnote)

- Analyze both binary presence/absence and continuous/ordinal measures of extent, comparing patterns across both operationalizations
- Report effect sizes for both "any exploratory research" and "substantial exploratory research" (e.g., >25% of results) to distinguish restriction from minimization
- Conduct sensitivity analyses examining whether patterns differ when restricting to articles with substantial versus minimal exploratory work
- Example expansion: "Code proportion of Results devoted to exploratory analyses: 0 (none), 1 (1-10%, typically single robustness check), 2 (11-25%, multiple supplementary analyses), 3 (26-50%, major results component), 4 (>50%, primary focus)"

1.2 Article Selection and Matching Procedures

Matching Hierarchy and Comparability -

Critique: The hierarchical matching prioritizing Timeframe over Topic/Design similarity requires reconsideration, as accepting Topic/Design ratings of 3 ("not similar") to maintain Timeframe 2 pairs fundamentally incomparable articles (e.g., depression interventions with cognitive neuroscience studies) from adjacent issues. This approach conflates registration effects with unmeasured confounds including statistical complexity, sample size requirements, field-specific norms regarding exploratory analysis, and disciplinary variations (experimental vs. observational traditions). The subjective 3-point ratings lack demonstrated reliability, no empirical justification supports prioritizing temporal proximity over substantive similarity, and no verification ensures balanced score distributions or sensitivity analyses excluding poor matches. Such mismatches could bias differences toward artifacts of topic clustering rather than registration status, particularly given that research areas adopting registration may differ systematically in complexity and analytic practices (Claesen et al., 2021).

Relevant section: "Article Selection Process" (pp.6-9), specifically the "Companion articles will be selected based on the following criteria" subsection describing matching hierarchy.

Framework for Addressing: - Reverse the hierarchy to prioritize Topic 2 and Design 2 as mandatory thresholds, then minimize Timeframe within these constraints

- Establish objective criteria for Topic/Design ratings with examples: Topic 1="both examine depression interventions," 2="both examine psychopathology," 3="both clinical but different domains"; provide similar specifications for Design
- Conduct inter-rater reliability assessment on Topic/Design ratings using 50 pilot articles, requiring $\kappa \geq .70$ before applying to full sample

- Report distributions of matching quality scores and conduct sensitivity analyses: (a) excluding matches with Topic or Design=3, (b) including matching quality as covariates in analyses
- Add statistical controls for measured confounds: code article complexity (number of studies, analyses, sample size categories) and research domain (experimental, observational, clinical, social, cognitive), include as covariates
- Example addition to matching criteria: "Companion articles must achieve minimum scores of 2 for both Topic and Design. Within this constraint, select the article with the closest publication date. If multiple articles meet all criteria, prioritize Topic similarity (1>2) over Design similarity."

Selection Bias From Sampling Frame -

Critique: The sampling frame introduces fundamental selection bias that cannot be separated from registration effects. Starting exclusively with 47 journals publishing Registered Reports means comparing traditional articles in journals that adopted RRs against traditional articles in those same journals. Journals adopting RRs likely possess pre-existing editorial cultures valuing transparency, exploratory research, and open science differently than psychology journals generally (Chambers & Tzavella, 2020). Any observed differences could reflect journal selection effects rather than registration effects—higher exploratory rates in registered articles might indicate (a) registration clarifying exploration, (b) RR-publishing journals attracting authors valuing exploration, or (c) editorial cultures encouraging exploratory reporting regardless of registration. The design provides no mechanism to distinguish these explanations.

Relevant section: "Article Selection Process" (pp.6-9), specifically the rationale "These journals are a good starting point, because publishing Registered Reports is a strong signal that the journal values registration."

Framework for Addressing: - Add a comparison group: sample traditional articles from journals that do not publish Registered Reports but match on impact factor, subdiscipline, and editorial scope to assess whether traditional articles in RR-publishing journals are representative

- Conduct supplementary analyses comparing traditional articles in RR-publishing journals to this matched control group from non-RR journals
- If adding this comparison is infeasible, acknowledge this as a critical limitation: "Our findings compare registered and traditional articles within journals adopting Registered Reports; results may not generalize to psychology broadly if these journals have distinctive editorial cultures"
- Code and analyze journal-level characteristics (e.g., open science policies, impact factor, publication volume) as potential moderators
- Frame conclusions specifically: "Within journals that publish Registered Reports..." rather than making general claims about registration's impact on psychology

Treatment of Multi-Study Articles and Unit of Analysis -

Critique: Including articles where only one study among multiple is registered while coding exploratory research article-wide introduces misclassification bias that obscures registration effects. Articles classified as "registered" may contain substantial exploratory work in non-registered studies, making them functionally indistinguishable from traditional articles regarding the non-registered portions. The justification—that researchers using mixed formats to include both confirmatory and exploratory analyses would be "consistent with registration advocates' position"—embeds the conclusion being tested into the methodological choice, creating

circular reasoning. More critically, this decision misaligns measurement with the theoretical level at which registration operates: registrations apply to specific studies, not entire articles. If authors strategically segregate exploration to non-registered studies within mixed articles, this would support rather than refute restriction concerns.

Relevant section: "Article Coding" (pp.9-10), particularly the paragraph stating "Coding will be done at the article-level rather than individual studies within articles."

Framework for Addressing: - Restructure to study-level analysis: for multi-study articles, code each study separately, recording whether that specific study was registered and the exploratory research within that study's corresponding results

- Compare registered studies to non-registered studies both between and within articles, allowing detection of whether authors segregate exploration by registration status
- If article-level coding is retained, restrict inclusion to articles where ALL studies share the same registration status (all registered or all non-registered) to ensure clean categorization
- Conduct sensitivity analyses comparing results when including versus excluding mixed-registration articles
- Code registration proportion for multi-study articles (e.g., 1 of 3 studies registered = 33%) and examine this as a continuous moderator to test dose-response relationships
- Report separately: analyses of pure-type articles only (all studies registered/non-registered) versus all articles including mixed types, with explicit comparison of conclusions

1.3 Coding Reliability and Measurement Quality

Adequacy of Reliability Thresholds -

Critique: The target thresholds (80% agreement, $\kappa.60$) require reconsideration given the subjective judgments required and the study's validity dependence on accurate identification. Kappa of .60 represents "moderate agreement" by conventional standards (Landis & Koch, 1977), yet coding involves inferring author intent through "clearly implied" meanings where exploratory research "lacks clear or consistent definition." In content analysis requiring subjective interpretation, standards typically mandate $\kappa.70$ -80 for acceptable reliability (Hallgren, 2012). Additionally, 80% agreement could mask systematic disagreements—if two raters consistently disagree on 20% of articles in the same direction (one systematically more lenient), this meets the threshold while introducing substantial bias. Given the hypothesized moderate effect ($\phi=.18$), measurement error at this level could produce spurious findings or obscure true effects.

Relevant section: "Rater Training and Reliability" (pp.11-12), specifically "If percent agreement exceeds .80 and Kappa exceeds .60, then the training phase will be complete."

Framework for Addressing: - Increase thresholds to 85% agreement and $\kappa.75$ before proceeding to focal coding, consistent with standards for subjective content analysis requiring inference about author intent

- Examine not only overall agreement but also direction of disagreements: calculate separate agreement for "exploratory present" versus "exploratory absent" decisions to detect systematic leniency/strictness biases
- Track disagreement patterns by article type during training to identify whether certain article types (e.g., traditional) are systematically more difficult to code

- If increased thresholds cannot be achieved after extended training, conduct all focal coding with three raters per article (instead of two), using majority rule or requiring consensus
- Report reliability metrics separately for each section (Introduction, Results, Discussion) and each article type, as reliability may vary across these dimensions
- Example addition: "During training, we will examine whether disagreements are randomly distributed or systematically favor certain coding decisions (e.g., one rater coding ambiguous cases as exploratory more often), adjusting training protocols if systematic biases emerge."

Blinking Procedures and Expectancy Effects -

Critique: The absence of blinding procedures in coding where raters identify "clearly implied" exploratory research requires attention, as knowledge of article type may introduce expectancy effects. Registered Reports' distinctive two-stage structure and preregistration statements make article types readily identifiable, potentially leading raters to apply differential thresholds when inferring unlabeled exploration: stricter thresholds for registered formats (expecting explicit labels per requirements) and looser ones for traditional articles. Training achieving only moderate reliability ($\kappa=.60$) provides insufficient safeguard against shared biases among the four-person team who collaborate weekly. No procedures for partial blinding (e.g., redacting registration information, removing Stage 1/2 headers), bias assessment (testing whether raters can identify article types and whether this affects coding), or differential error checks (comparing disagreement rates by article type) are specified.

Relevant section: "Rater Training and Reliability" (pp.11-12) and "Article Coding" (pp.9-11); refers to omissions throughout.

Framework for Addressing: - Implement partial blinding: create redacted versions removing obvious registration indicators (preregistration badges, Stage 1/Stage 2 section headers, explicit registration statements, OSF links) for a subset of articles

- Randomly assign 100 articles to blinded condition (redacted) and 200 to unblinded condition; compare coding patterns across conditions to assess bias magnitude
- Add explicit bias checks: after completing 50% of coding, have raters independently guess each article's type; calculate identification accuracy and test whether articles where type was correctly identified show different coding patterns
- Include "confidence in coding decision" ratings (1-5 scale) to identify cases where rater uncertainty might reflect ambiguity rather than true construct presence
- Analyze inter-rater reliability separately by article type; substantially lower reliability for one type would indicate differential measurement quality requiring investigation
- Example: "To assess potential expectancy effects, we will redact registration-identifying information from 100 randomly selected articles and compare coding reliability and outcomes between redacted and non-redacted versions."

1.4 Statistical Approach and Power

Accounting for Nested Data Structure -

Critique: The planned chi-square tests require reconsideration given multiple non-independence sources violating the tests' assumptions. Articles are explicitly matched by journal (creating clustering within journals sharing editorial policies, reviewer pools, and cultures), may originate

from identical research groups, and exhibit temporal clustering (same/adjacent issues). The matching procedure intentionally creates dependencies: companion articles selected from identical journals and proximate issues means observations are nested within journals and time periods. Chi-square assumes independent observations; these violations will inflate Type I error rates and bias effect size estimates (McNeish & Stapleton, 2016). The analysis plan mentions neither hierarchical/multilevel modeling to accommodate clustering, nor sensitivity analyses for independence violations, nor corrections for non-independence.

Relevant section: "Planned Analysis" (pp.12-13), particularly "Hypothesis Tests" subsection specifying "2x2 chi-square test of independence ($\alpha = .05$)" and "3x3 chi-square test of independence."

Framework for Addressing: - Replace chi-square tests with generalized linear mixed models (GLMMs) specifying random intercepts for journal and potentially time period (publication year or issue number)

- For binary outcomes (exploratory research present/absent), use mixed-effects logistic regression; for averaged ratings, use linear mixed models
- Report intraclass correlation coefficients (ICCs) quantifying the proportion of variance attributable to journal-level versus article-level factors
- Conduct sensitivity analyses comparing naive chi-square results to mixed model results to demonstrate the impact of accounting for clustering
- If sample size per cluster is small (few articles per journal), consider cluster-robust standard errors as alternative approach
- Example model specification: "We will fit a mixed-effects logistic regression predicting exploratory research presence from article type (RR, preregistered, traditional), with random intercepts for journal to account for clustering. Fixed effects will be tested using Wald tests with Kenward-Roger degrees of freedom adjustment."

Multiple Comparisons and Family-Wise Error -

Critique: The analysis plan requires addressing multiple comparisons despite planning numerous hypothesis tests, which substantially inflates family-wise Type I error rate. Planned analyses include: (1) 2x2 chi-square for H1, (2) "secondary test" examining headers for H1, (3) 3x3 chi-square for H2, (4) "secondary test" examining headers for H2, and (5) follow-up cell-wise residual analyses for significant results using ± 1.96 thresholds—minimally 4-6 inferential tests plus additional follow-ups if omnibus tests achieve significance. The "primary" versus "secondary" distinction lacks statistical meaning for family-wise error control; both contribute to inflation. With 4-6 planned tests at $\alpha = .05$, probability of 1 Type I error under the null substantially exceeds the nominal .05 level. The proposal's emphasis on providing "strong data" for policy discussions heightens the importance of controlling false positive rates (Rothman, 1990) (Benjamini & Hochberg, 1995).

Relevant section: "Planned Analysis" (pp.12-13), "Hypothesis Tests" subsection; refers to omission of multiple comparison corrections throughout.

Framework for Addressing: - Specify a priori which tests constitute the primary confirmatory family requiring correction (minimally: the four hypothesis tests for H1 and H2, both primary and secondary)

- Apply Holm-Bonferroni sequential procedure or FDR correction (Benjamini-Hochberg) to this

family, adjusting the decision rule accordingly

- For follow-up tests (cell-wise residuals), apply correction within each omnibus test family or use more stringent thresholds (e.g., ± 2.58 for $p < .01$)
- Distinguish clearly between confirmatory tests (corrected α) and truly exploratory analyses (reported without p-values as planned)
- Report both uncorrected and corrected results for transparency, but base conclusions on corrected values
- Example: "We will control family-wise error rate across the four primary hypothesis tests using the Holm-Bonferroni method. For H1 and H2 to be considered supported, both the primary and secondary tests must achieve significance at the adjusted α levels. Follow-up analyses will be conducted only for omnibus tests surviving correction."

1.5 Confounds and Alternative Explanations

Temporal Confounding With Historical Comparison Group -

Critique: The inclusion of 75 traditional articles from 2010 "prior to registration adoption" introduces temporal confounding that compromises primary analyses. Comparing recent registered articles (2015-2026) to a mixed pool including 2010 articles creates ambiguity: observed differences could reflect registration effects, 15-year secular changes in reporting norms, or their interaction. This period witnessed substantial evolution: the replication crisis gained prominence (2011-2012), transparency initiatives proliferated, APA reporting standards updated (2010, 2013, 2020), and journal policies regarding open science badges and data sharing increased dramatically—affecting traditional articles independently of preregistration. The power analysis assumes simple comparisons between contemporaneous groups, but the design involves more complex structure where one category spans different (and longer) timeframes. The current approach confounds factors in primary analyses while relegating time investigation to "exploratory" analyses without inferential tests.

Relevant section: "Article Selection Process" (pp.6-9), particularly the paragraph describing "a second group of articles published in 2010, prior to the adoption of registration in psychology."

Framework for Addressing: - Restrict all articles to the same time window (2015-2026 only) for primary hypothesis tests, ensuring temporal comparability

- Analyze 2010 articles separately as a distinct historical baseline, not combined with matched traditional articles in primary analyses
- If 2010 articles are retained, explicitly model time as a factor: use $2 \times 2 \times 2$ design (Article Type: Registered/Traditional $\times$ Time Period: 2010/2015-2026) allowing separation of registration and temporal effects
- Test registration $\times$ time interaction to determine whether registration's association with exploratory reporting differs between periods
- Report primary analyses both with and without 2010 articles, clearly labeling which constitute the definitive hypothesis tests
- Example revision: "Primary analyses will compare 75 Registered Reports, 75 preregistered articles, and 75 matched traditional articles all from 2015-2026. Separately, we will compare the 75 matched traditional articles (2015-2026) to 75 traditional articles from 2010 to assess temporal changes in reporting practices independent of registration."

Registration Quality as Unmeasured Confounder -

Critique: The decision to remain "agnostic about registration quality" constitutes a critical design limitation given documented variation in registration specificity. Multiple studies demonstrate enormous quality heterogeneity (van den Akker et al., 2023), with poor-quality registrations lacking specificity functionally rendering all analyses exploratory by default—artificially inflating exploratory reporting in registered articles. Conversely, high-quality registrations with specific predictions might show reduced exploratory work as critics predict, but this effect would be masked by including poor registrations in the same category. The authors acknowledge this variation exists but dismiss it as "not our concern here"—yet if registration quality is the actual causal mechanism (only high-quality registrations potentially restricting exploration), ignoring it violates the requirement of measuring treatment actually delivered. The power analysis assumes homogeneous treatment effects; unmeasured heterogeneity could mean the study lacks power to detect effects within quality strata.

Relevant section: "Article Coding" (pp.9-11), paragraph stating "in the current project we remain agnostic about the quality of the registrations themselves."

Framework for Addressing: - Code registration specificity for all registered articles using established frameworks: specify whether hypotheses, predictions, analysis plans, and exclusion criteria are clearly detailed (binary or 4-point scale per element)

- Calculate overall registration quality score (e.g., sum across elements, range 0-4 or 0-8) and categorize as high/medium/low quality using quartiles or established cutoffs

- Conduct analyses stratified by registration quality: test whether associations between registration and exploratory reporting differ across quality levels

- Report separate effect estimates for high-quality versus low-quality registrations to determine if patterns are quality-dependent

- Test registration quality as a moderator using interaction terms in statistical models

- Example addition: "We will assess registration quality by coding four elements: (1) specificity of hypotheses/predictions, (2) detail of planned analyses, (3) specification of exclusion criteria, (4) clarity of outcomes (each rated 0-2). We will examine whether associations between registration status and exploratory reporting vary by total quality score (0-8), testing whether effects are driven by high-quality registrations only."

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Appendix 1: Most Similar Publications